

# Bayesian Network Modeling of NFL Game Stats & Market Signals

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**3,990**

NFL Games

**15**

Seasons

**6**

Market Pts / Game

**2**

Data Sources

# The Question, Before the Math

Does any of this tell you something the box score doesn't?

## ● NFL · CLOSING LINES

illustrative

| MATCHUP    | SPREAD (OPEN → CLOSE) |   |             | TOTAL | MONEYLINE |
|------------|-----------------------|---|-------------|-------|-----------|
| IND vs NO  | -5.0                  | → | -5.5        | 52.5  | -240      |
| PIT vs CLE | -4.5                  | → | -5.0        | 36.5  | -235      |
| GB vs PHI  | -2.5                  | → | <b>-3.0</b> | 43.0  | -150      |
| DEN vs BUF | -3.0                  | → | -2.5        | 38.5  | -130      |

Opening line → closing line. The **movement** is the signal we test against the box score.

# Objectives

Is line movement **conditionally independent** of ATS outcome given game fundamentals?

## Objectives

- Test whether line movement is conditionally independent of ATS outcome given game fundamentals
- Isolate whether market signals add predictive information beyond game statistics

Our models proved that **market signals**:

- **do NOT** improve accuracy beyond game stats
- **but they DO** improve calibration

# Gap in the Literature

## BNs for Sports Prediction

Constantinou, Fenton & Neil (2012–13): BN for soccer forecasting — game variables only, no market movement.

## Stats + Odds Combined

Egidi, Pauli & Torelli (2018): hierarchical Bayes blending odds — no conditional dependence structure.

## Market Efficiency

Simon (2024), Moskowitz (2021): line-movement efficiency — no game-stats integration.

**GAP CONFIRMED**

# Methodology

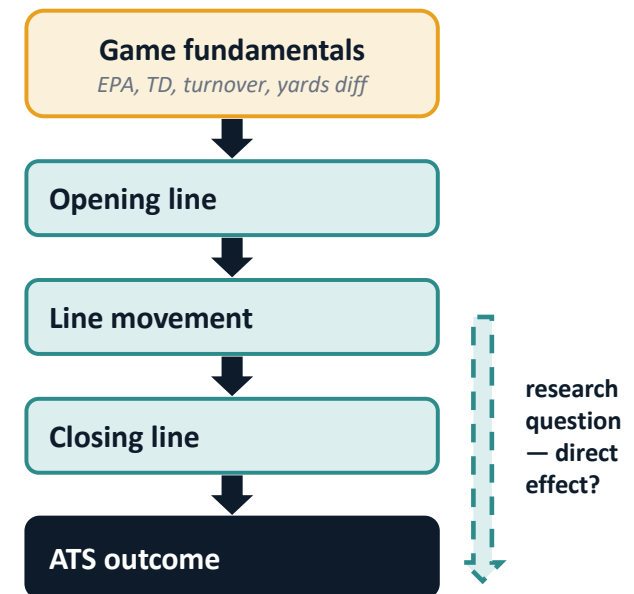
## MODEL ARCHITECTURE

- Naive Bayes baseline — all features independent
- Hill Climbing BN — score-based learning
- PC Algorithm BN — constraint-based check

## EVALUATION

- Accuracy — hard prediction performance
- Log Loss — probability calibration
- Target: Cover / Loss / Push (3-class)

## ASSUMED CAUSAL STRUCTURE



Ablation: 3 feature sets (Stats · Market · Combined) × 3 models (NB · HC · PC) = 9 configurations, each scored on accuracy and log loss.

# The Result Is Not What You'd Expect

## ACCURACY

Market signals **HURT**

**84.1%**

Stats-only Naive Bayes

**81.0%**

Combined (stats + market)

Adding market features lowers accuracy.

vs

## CALIBRATION

Market signals **HELP**

**0.345**

Combined PC log loss (best)

**0.439**

Stats-only NB log loss

Adding market features improves calibration.

# Results: Accuracy vs. Calibration

## ACCURACY

**84.1%**

Stats-only Naive Bayes is the highest of any configuration. Adding market features lowers NB accuracy to 81.0%; both Bayesian Networks stay at or below the 58.0% majority baseline in every configuration.

## CALIBRATION (LOG LOSS)

**0.345**

Combined PC algorithm — best log loss of all nine runs. Market signals sharpen probabilities without changing hard predictions.

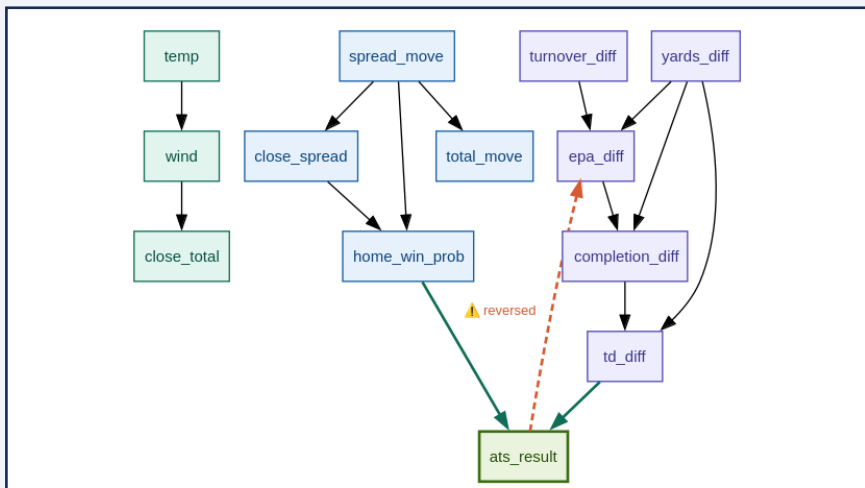
| Config      | NB Acc | HC Acc | PC Acc | NB Loss | HC Loss | PC Loss |
|-------------|--------|--------|--------|---------|---------|---------|
| Stats-Only  | 84.1%  | 50.4%  | 50.4%  | 0.439   | 0.375   | 0.368   |
| Market-Only | 56.4%  | 58.6%  | 58.6%  | 0.840   | 0.776   | 0.760   |
| Combined    | 81.0%  | 50.4%  | 47.3%  | 0.538   | 0.375   | 0.345   |

## CENTRAL FINDING

Market signals improve how certain the model is — not what it predicts.

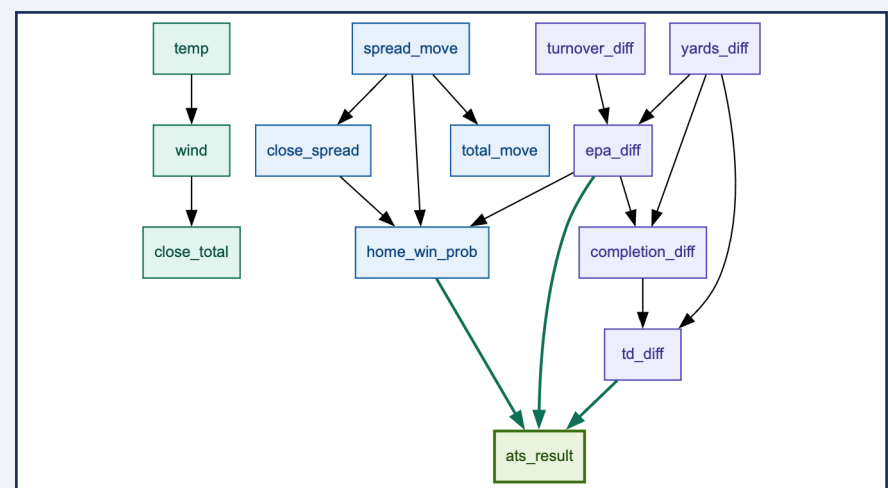
## The Model Made a Causal Error — a Second Algorithm Caught It

### HILL CLIMBING — raw learned DAG



Hill Climbing learned  $\text{ats\_result} \rightarrow \text{epa\_diff}$  — a betting outcome cannot cause the on-field play that produced it. Causally impossible.

### PC ALGORITHM — independent result



Run independently, the PC algorithm recovered  $\text{epa\_diff} \rightarrow \text{ats\_result}$  on its own — with  $\text{home\_win\_prob}$  and  $\text{close\_spread}$  as direct parents. Not confirmation bias: a second method found the same structure.

**The algorithm made a causal error — and we caught it:**

domain knowledge flipped the edge, and a second, independent algorithm (PC) recovered the same direction on its own.



# Implementation

## DATA PIPELINE

- nflverse (game stats) + SportsBookReviewsOnline (market data)
- 3,990 games, 2007–2022, joined by season + matchup
- Game Stats: EPA, TD, turnover, yards differentials
- Market Signals: spread & total movement, home win prob, rolling ATS

## MODELS

- Gaussian Naive Bayes => continuous features, no discretization
- Hill Climbing BN + PC Algorithm BN => 3-bin ordinal encoding
- 80/20 train-test split

## VALIDATION

- Dual-algorithm approach: HC and PC run independently and cross-checked
- HC reversed edge corrected via domain knowledge, confirmed by PC

# Challenges & Limitations

## Retrospective Statistics

EPA and other game stats are post-game measurements. Stats-only accuracy reflects information content, not pre-game predictive power.

## Discretization Penalty

BNs require binning continuous features; magnitude information is lost. The main reason BNs trail Naive Bayes on accuracy.

## Class Imbalance

Push is only 3.0% of the data. Neither BN ever predicts it; NB reaches just 22% F1 on the class.

## Spread Direction

We bin the closing spread by magnitude, which discards direction: a home team laying seven and one receiving seven fall in the same bin. Recovering signed direction is left to future work.

# Future Work & Conclusions

## FUTURE WORK

- Continuous (Gaussian) BN formulations to eliminate the discretization penalty
- Post-2018 PASPA analysis only — isolate the modern legal betting market
- Recover spread direction; revalidate the target variable
- Add quarterback injury data as a high-signal feature

## CONCLUSIONS

- Partial efficiency confirmed: the market's hard prediction is already encoded in game stats
- Calibration finding: the market's uncertainty estimate adds information beyond statistics alone
- PC + HC dual-algorithm design caught and corrected a causally impossible edge
- A nuanced answer that advances both sides of the efficiency debate

## QUESTIONS & DISCUSSION

**For accuracy: the market adds no signal beyond game stats.**

**For calibration: the market's uncertainty estimate does.**

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